Technical Report

Objective:

To understand if price affects customer satisfaction and whether most engagement comes from only a few products.

Data Preparation

1. Load the Dataset:
    - Imported dataset using Pandas.
    - Loaded CSV file into DataFrame.
    - Inspected the initial structure, such as the length of the DataFrame, data types, and structure of the DataFrame.

2. Data types correction and standardisation:
    - Cleaned several numeric columns that were incorrectly stored as objects due to special characters.
    - Removed unnecessary symbols and formatting characters.
    - Converted string to numeric in some attributes.
    - Ensured numerical columns are properly formatted for further analysis.
    - This step improved **data integrity and analytical usability**.

3. Identifying Numerical vs Categorical Variables:
    - Understand feature
    - Prepare for Exploratory Data Analysis

4. Handling duplicates:
    - Duplicate analysis revealed no exact row duplication.
    - Some columns showed repetition due to natural one-to-many relationships in data.

5. Removing Irrelevant Columns:
    - Dropped non-analytical columns.
    - Do not contribute to structured price/rating analysis.
    - Are mostly textual or identifiers.
    - Increase dimensionality unnecessarily.

6. Feature Engineering:
    - The category column contained unnecessary values that were hard to interpret and difficult to analyse.
    - Split the column into a hierarchical structure for better understanding.
        i. Root (Department)
        ii. Category
        iii. Sub-Category
        iv. Sub-Sub Category
        v. Product type

- Better hierarchical analysis
- Department-wise insights
- Category-level performance tracking
- More interpretable business analysis
- This is proper feature engineering aligned with Amazon's product taxonomy.

7. Handling Missing values:
    - Identified missing values and found them only in rating and rating-count.
    - Instead of immediately filling values, you:
        a. Investigated the rows where missing values existed.
        b. Observed an important pattern:
        When the rating was missing, and the rating-count existed.
        When the rating-count was missing, and the rating existed.
    - Cleaned the missing values according to the group logical relationship.
    - Mode used if the missing value is random.
    - If columns are related (e.g. Category and Sub-category), group-wise mode is better.
    - If dependency exists, use conditional imputation instead of global mode.

8. Saved the file into CSV.

Exploratory Data Analysis

1. **Univariate Analysis**
   The purpose of performing Exploratory Data Analysis (EDA) was to:
   - Understand the distribution of each numerical variable.
   - Identify central tendency and variability.
   - Detect skewness and outliers.
   - Determine appropriate statistical measures for interpretation.
   - Establish a statistical foundation before moving to relationship analysis.

a. Descriptive Statistics
    - Generated summary statistics.
    - Compared mean and median to detect the skewness.
    - Analysed spread using standard deviation and IQR.
    - Evaluated min–max range to detect extreme dispersion.

    Observation:
    - Actual-price and discounted-price: Mean significantly greater than Median (Right-skewed).
    - Rating-count: Highly right-skewed.
    - Discount-percentage: Mean approximately equal to Median (Symmetric).
    - rating: Slightly concentrated around the mean.

b. Distribution Analysis
    Observation:
    From the histogram plot
    - Confirm skewness

- Identify long-tail behaviour

From the Boxplot visuals

- Visualise IQR
- Confirm presence of outliers

c. Outlier detection through IQR method
Identified number of outliers per variable:
- Discounted-price: 217 outliers
- Actual-price: 213 outliers
- Rating: 74 outliers
- Rating-count: 140 outliers
- Discount-percentage: 0 outliers

Key Observation:
There is high variability relative to central tendency.

2. Bivariate Analysis

The purpose of performing Bivariate Analysis was to:
- Examine relationships between numerical variables.
- Identify strength and direction of associations.
- Understand whether pricing, discounting, and engagement variables are related.
- Evaluate the impact of skewness and outliers on relationships.
- Derive business insights before proceeding to SQL-based analytical queries.

a. Numerical vs Numerical Analysis (Correlation Analysis)
- Computed Pearson correlation to measure linear relationships.
- Computed Spearman correlation to account for skewness and outliers.
- Compared both methods to ensure robustness of findings.

Observation:
- No strong linear relationships were observed across most numerical pairs.
- Discount-percentage showed weak association with rating and rating-count.
- Rating and rating-count displayed limited monotonic association.
- Actual-price and discounted-price showed strong positive association (structurally dependent).

Most pricing and discounting variables do not exhibit strong direct statistical dependency with customer rating behaviour, suggesting the presence of additional influencing factors such as category or product type.

b. Scatter Plot Analysis
- Generated scatter plots for meaningful variable pairs.
- Applied log transformation to skewed variables (actual-price, discounted-price, rating-count) for visual clarity.
- Compared patterns before and after transformation.

Observation:

- No clear strong linear pattern emerged in most cases.
- Relationship patterns appeared dispersed, indicating weak predictive dependency.
- Log transformation reduced visual compression caused by extreme values.

Solving Queries

After completing data cleaning and exploratory data analysis, the dataset was imported into SQLite to:

- Organise data into structured relational format
- Enable aggregation-based analysis
- Create reusable business-level tables

Solved 12 queries that were designed to measure engagement concentration, pricing patterns, statistical validity of ratings, revenue proxy estimation, and product segmentation.

SQL concepts which are performed:

AGREGATE FUNCTION (COUNT, AVG, SUM, etc.)

CONDITIONS

SUBQUERIES

WINDOW FUNCTIONS (RANK, OVER..)

CTES (Common Table Expression)

1. **Category-Level Aggregation**
   To analyse engagement concentration at the category level, a new table was created.

   - Total Products
   - Total Engagement
   - Engagement per Product
   - Average Final Price
   - Revenue Proxy (Rating-count was considered as Quantity sold on the temporary basis)
   - Engagement Share percentage

2. **Product-level Aggregation**
   The product table was created for detailed performance analysis.
   - Price-segment: To know the range of the product is in Budget, Mid-Range or Premium.

- Engagement-level: To analyse the most engaging product in the range of high, medium or low.
- Performance Segment: To know the performance of the product with ratings and the rating-count highlighting in Star product, Hidden Gem or Mass but Risky category.

Observation:

Can see the impact of skewness and outliers.

Treated revenue as engagement-weighted proxy.

Dashboards

Created an interactive, business visualisation dashboard which helped to understand the customers' behaviour towards the products, categories by rating them according to their experience towards the product, and to represent the segments' growth or risk.

1. Hierarchical Aggregation Strategy
   The dataset contained a multi-level product taxonomy structured as:
   - Department
   - Category
   - Sub-Category
   - Sub-Sub Category
   - Product Type

   The full hierarchy was constructed and retained within the data model, only selected levele were used for visualisation, like Category for strategic-level to understand which categories are dominating, and the other is Product type for operational level. Department-level provides macro-level concentration insight.

   A selective hierarchical strategy was adopted to balance analytical depth and business readability.

Key Measures (KPIs) created in DAX:

- Total Engagement
- Weighted Average Rating
- Engagement per Product
- Engagement-Weighted Revenue (Proxy)

Drill-Down Implementation

- Hierarchy created:

- Department
  - Category
  - Product_type

Limitations:

The dataset does not contain actual sales volume or revenue figures. Rating-count was used as a proxy for customer engagement and approximate demand, which may not fully represent true purchasing behaviour.

Several numerical variables (price and rating-count) exhibited right-skewed distributions with significant outliers. Although transformation was used for analytical validation, dashboard metrics are based on raw values, which may slightly influence aggregation results.

Weak correlations between the variables.

Recommendations:

The company should review its discount strategy because higher discounts do not always lead to more engagement or customer satisfaction.

High-engagement but low-rated products should be prioritized for quality improvement to prevent long-term brand perception damage.